

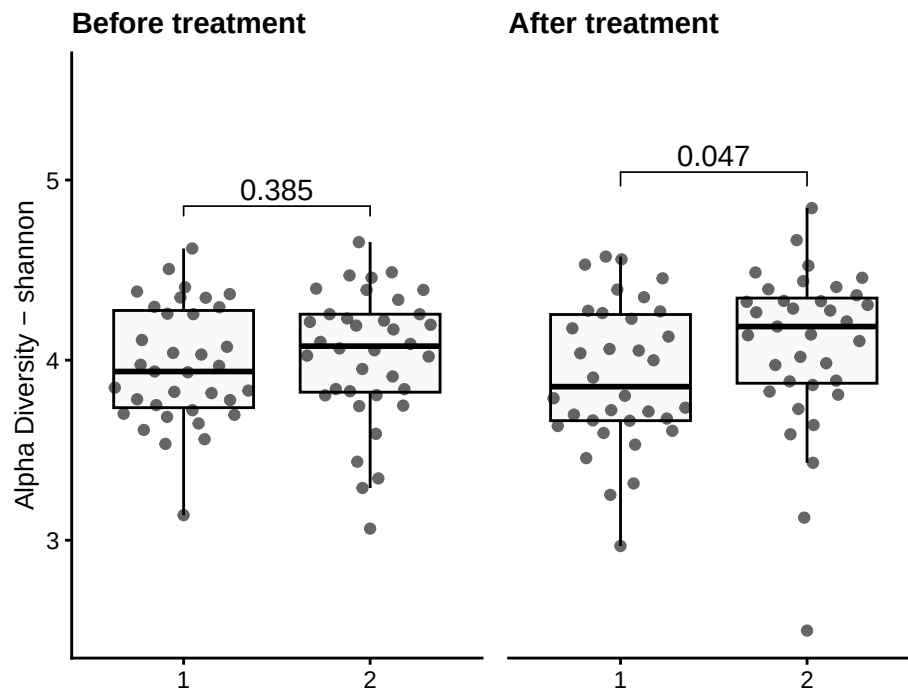
Alpha Diversity

Muluh

1 Community diversity

2 Group-wise comparisons

- **Index:** shannon
- **Group variable:** diet
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon



2.1 Table of the top findings

2.1.1 Across-group

2.1.1.1 Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	4	4	0.02	0.385

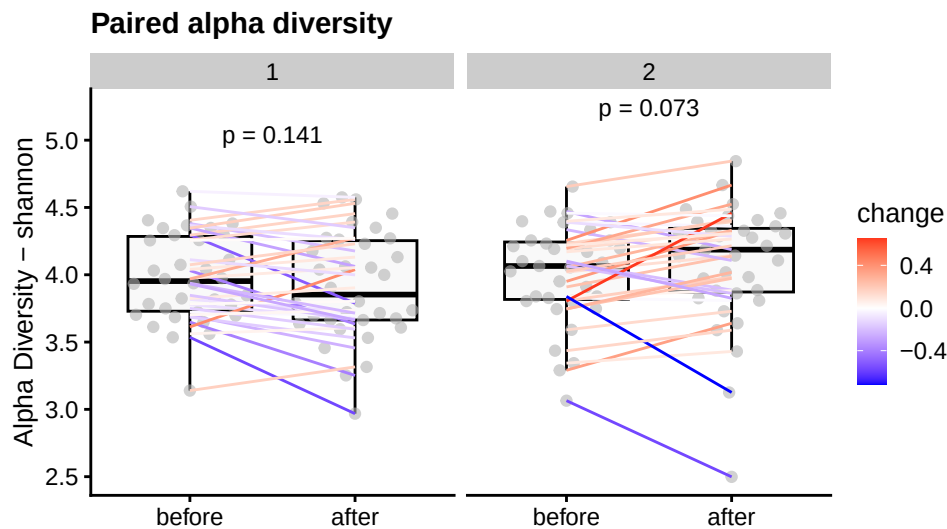
2.1.1.2 After treatment

- Multiple testing correction: **fdr**

- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	3.9	4.1	0.06	0.047



2.1.2 Within-group

2.1.2.1 Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value
1	4	3.9	-0.02	0.141
2	4	4.1	0.03	0.073

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: y ~ diet * time + (1 | id)

Data: df_no_miss

REML criterion at convergence: 75.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.62184	-0.44101	0.01695	0.47507	1.84121

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.11992	0.3463
Residual		0.03086	0.1757

Number of obs: 140, groups: id, 71

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	4.03642	0.08878	135.88442	45.468	<2e-16 ***

Characteristic	before			after		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	3.6	3.1, 4.0	<0.001	3.5	2.9, 4.1	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.21	-0.03, 0.44	0.084	0.28	-0.02, 0.58	0.065
3	0.12	-0.12, 0.35	0.3	0.20	-0.09, 0.50	0.2
4	0.17	-0.06, 0.40	0.2	0.10	-0.19, 0.39	0.5
age	0.01	0.00, 0.01	0.2	0.01	0.00, 0.02	0.2

¹CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI ¹	p-value
(Intercept)	3.5	-Inf, Inf	<0.001
meal_group			0.2
1	—	—	
2	0.23	-Inf, Inf	>0.9
3	0.16	-Inf, Inf	>0.9
4	0.13	-0.05, 0.32	0.2
age	0.01	0.00, 0.01	0.039

¹CI = Confidence Interval

```
diet2      -0.07679    0.12467 135.88494  -0.616    0.5389
time       -0.06274    0.04254  67.44785  -1.475    0.1450
diet2:time  0.12717    0.05973  67.44297   2.129    0.0369 *
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Correlation of Fixed Effects:
      (Intr) diet2  time
diet2    -0.712
time     -0.713  0.507
diet2:time 0.508 -0.713 -0.712
```

3 Alternate analyses

- **Beta (Estimate):** Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI:** Range where the true beta is likely to fall, reflecting precision.
- **p-Value:** Tests if beta differs from zero; $p < 0.05$ suggests significance.

4 Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
```

```
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(14): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(14): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```